MIPT & YSDA · AUTUMN 2026

# Deep Generative Models

## Lecture 6

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# Recap of Previous Lecture

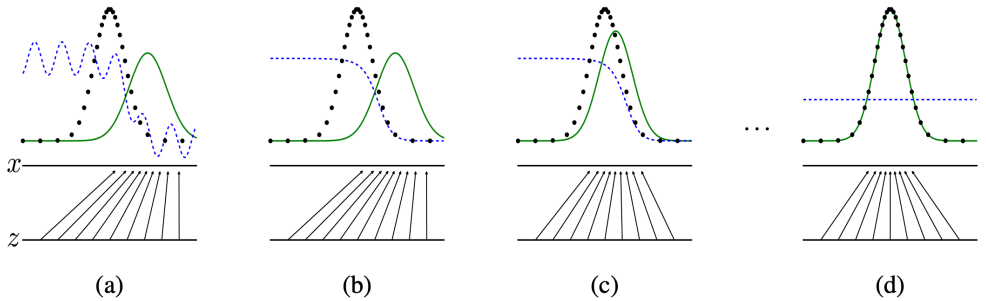
## Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))) \right]$$

## Optimal Discriminator and Jensen-Shannon Divergence

For fixed $G$, write the bracketed objective as $V(G, D)$:

$$D^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_{\boldsymbol{\theta}}(\mathbf{x})}, \qquad V(G, D^*) = 2\, D_{\text{JS}}(p_{\text{data}}(\mathbf{x}) \,\|\, p_{\boldsymbol{\theta}}(\mathbf{x})) - 2 \log 2.$$




Mode collapse refers to the phenomenon where the generator in a GAN produces only one or a few different modes of the distribution.

Goodfellow I. J. et al. Generative Adversarial Networks, 2014
Metz L. et al. Unrolled Generative Adversarial Networks, 2016

# Recap of Previous Lecture

GAN losses also train the reconstruction stage of a **latent generative model**.

## Stage 1: Train the Autoencoder

A fixed pretrained network $F$ defines the perceptual loss for reconstruction $\hat{\mathbf{x}}$:

$$\mathcal{L}_{\text{perc}} = \|F(\mathbf{x}) - F(\hat{\mathbf{x}})\|_2^2.$$

Train a discriminator $D$ to separate real and reconstructed image patches; train the autoencoder to fool $D$.

A KL-autoencoder also uses weak KL regularization.

## Stage 2: Freeze the Autoencoder and Learn a Prior

Fit the distribution of its latent codes — the **aggregated posterior**.

**VQGAN:** sample a code map $\mathbf{c}$ from an AR prior, look up its embeddings $\mathbf{z}_q(\mathbf{c})$, then decode.

$$\mathbf{c} \sim p_{\psi}(\mathbf{c}) \;\longrightarrow\; \mathbf{z}_q(\mathbf{c}) \;\longrightarrow\; \hat{\mathbf{x}}.$$

**KL-autoencoder:** a diffusion prior over continuous latents.

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \;\longrightarrow\; \mathbf{z} \sim p_{\psi}(\mathbf{z}) \;\longrightarrow\; \hat{\mathbf{x}}.$$

Esser P., Rombach R., Ommer B. Taming Transformers for High-Resolution Image Synthesis, 2021
Rombach R. et al. High-Resolution Image Synthesis with Latent Diffusion Models, 2022

# Recap of Previous Lecture

## Standard GAN Objective

$$\min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi}} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log D_{\boldsymbol{\phi}}(\mathbf{x}) + \mathbb{E}_{p(\mathbf{z})} \log(1 - D_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z}))) \right]$$

## Theoretical Results

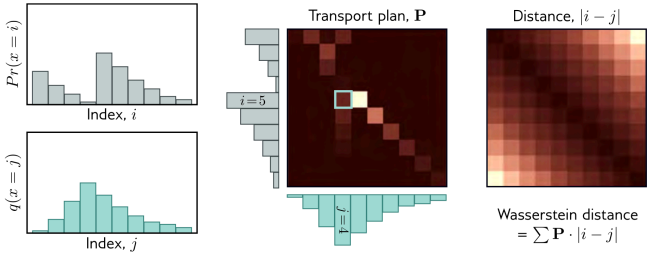
- Both the data distribution $p_{\text{data}}(\mathbf{x})$ and the generative distribution $p_{\boldsymbol{\theta}}(\mathbf{x})$ are supported on low-dimensional manifolds.
- If $p_{\text{data}}(\mathbf{x})$ and $p_{\boldsymbol{\theta}}(\mathbf{x})$ are disjoint, a smooth optimal discriminator can exist!

For such low-dimensional, disjoint manifolds:

$$D_{\text{KL}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = D_{\text{KL}}(p_{\boldsymbol{\theta}} \,\|\, p_{\text{data}}) = \infty, \qquad D_{\text{JS}}(p_{\text{data}} \,\|\, p_{\boldsymbol{\theta}}) = \log 2$$

Weng L. From GAN to WGAN, 2019
Arjovsky M., Bottou L. Towards Principled Methods for Training Generative Adversarial Networks, 2017

# Recap of Previous Lecture




## Wasserstein Distance

$$W(\pi\|p) = \inf_{\gamma\in\Gamma(\pi,p)} \mathbb{E}_{(\mathbf{x}_1,\mathbf{x}_2)\sim\gamma}\|\mathbf{x}_1 - \mathbf{x}_2\|$$

$$= \inf_{\gamma\in\Gamma(\pi,p)} \int \|\mathbf{x}_1 - \mathbf{x}_2\|\gamma(\mathbf{x}_1,\mathbf{x}_2)d\mathbf{x}_1 d\mathbf{x}_2$$

- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the transport plan: the amount of "dirt" assigned from $\mathbf{x}_1$ to $\mathbf{x}_2$.
- $\Gamma(\pi, p)$ denotes the set of all joint distributions $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ with marginals $\pi$ and $p$ ($\int \gamma(\mathbf{x}_1, \mathbf{x}_2)d\mathbf{x}_1 = p(\mathbf{x}_2)$, $\int \gamma(\mathbf{x}_1, \mathbf{x}_2)d\mathbf{x}_2 = \pi(\mathbf{x}_1)$).
- $\gamma(\mathbf{x}_1, \mathbf{x}_2)$ is the mass, $\|\mathbf{x}_1 - \mathbf{x}_2\|$ is the distance.

Simon J.D. Prince. Understanding Deep Learning, 2023
Arjovsky M., Chintala S., Bottou L. Wasserstein GAN, 2017

# Recap of Previous Lecture

On a compact data space $\mathcal{X}$:

## Dependence on Generator Parameters

For a feedforward generator $\mathbf{G}_{\boldsymbol{\theta}}$ with Lipschitz activations and $\mathbb{E}_{p(\mathbf{z})}\|\mathbf{z}\| < \infty$, $W(p_{\text{data}}\|p_{\boldsymbol{\theta}})$ is continuous in $\boldsymbol{\theta}$ and differentiable almost everywhere.

## Convergence of Distributions

For a fixed distribution $\pi$ and a sequence $p_t$ on $\mathcal{X}$, as $t \to \infty$:

$$D_{\text{KL}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad D_{\text{JS}}(\pi\|p_t) \to 0 \quad \Longrightarrow \quad W(\pi\|p_t) \to 0$$

The first implication also holds for $D_{\text{KL}}(p_t\|\pi) \to 0$; the converses need not hold.

> **Wasserstein can capture distributions getting closer even when their supports do not overlap.**

# Recap of Previous Lecture

### Theorem (Kantorovich-Rubinstein Duality)

$$W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) = \frac{1}{K} \max_{\|f\|_L \leq K} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right],$$

where $f : \mathbb{R}^m \to \mathbb{R}$ is $K$-Lipschitz continuous ($\|f\|_L \leq K$).

### WGAN Objective

$$\min_{\boldsymbol{\theta}} W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) \approx \min_{\boldsymbol{\theta}} \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p(\mathbf{z})} f_{\boldsymbol{\phi}}(\mathbf{G}_{\boldsymbol{\theta}}(\mathbf{z})) \right].$$

- The discriminator $D$ is replaced by function $f$: in WGAN, it is known as the **critic**, which is *not* a classifier.
- If the weights $\boldsymbol{\phi}$ are restricted to a compact set $\boldsymbol{\Phi} = [-c, c]^d$, then $f_{\boldsymbol{\phi}}(\mathbf{x})$ is $K$-Lipschitz continuous.

$$\begin{aligned} K \cdot W(p_{\text{data}} \| p_{\boldsymbol{\theta}}) &= \max_{\|f\|_L \leq K} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f(\mathbf{x}) \right] \\ &\geq \max_{\boldsymbol{\phi} \in \boldsymbol{\Phi}} \left[ \mathbb{E}_{p_{\text{data}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) - \mathbb{E}_{p_{\boldsymbol{\theta}}(\mathbf{x})} f_{\boldsymbol{\phi}}(\mathbf{x}) \right] \end{aligned}$$

# Recap of Previous Lecture

## Frechet Inception Distance (FID)

Use Inception image embeddings $\mathbf{z} = \mathbf{f}(\mathbf{x})$; compare Gaussian approximations of their distributions:

$$\begin{aligned}\mathrm{FID}(p_{\text{data}}, p_{\boldsymbol{\theta}}) = \|\boldsymbol{\mu}_{\text{data}} - \boldsymbol{\mu}_{\boldsymbol{\theta}}\|^2 \\ + \mathrm{tr}\left[\boldsymbol{\Sigma}_{\text{data}} + \boldsymbol{\Sigma}_{\boldsymbol{\theta}} - 2\left(\boldsymbol{\Sigma}_{\text{data}}^{1/2}\boldsymbol{\Sigma}_{\boldsymbol{\theta}}\boldsymbol{\Sigma}_{\text{data}}^{1/2}\right)^{1/2}\right].\end{aligned}$$

FID depends on the feature network, Gaussian approximation and sample size; it may not correlate with human evaluation.

## CMMD: Comparing CLIP Image Embeddings

For CLIP image-embedding distributions $P, Q$, use a Gaussian kernel $k$:

$$\mathrm{MMD}_k^2(P, Q) = \mathbb{E}k(\mathbf{z}, \mathbf{z}') + \mathbb{E}k(\mathbf{w}, \mathbf{w}') - 2\mathbb{E}k(\mathbf{z}, \mathbf{w}).$$

Here $\mathbf{z}, \mathbf{z}' \sim P$ and $\mathbf{w}, \mathbf{w}' \sim Q$ are mutually independent. This compares more than means and covariances.

Heusel M. et al. GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium, 2017
Jayasumana S. et al. Rethinking FID: Towards a Better Evaluation Metric for Image Generation, 2024

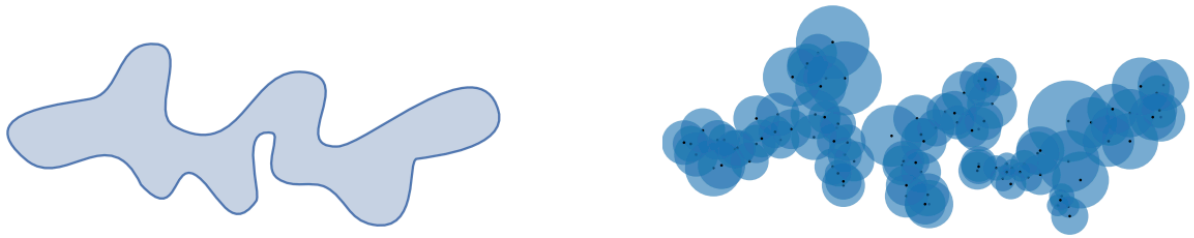
# Recap of Previous Lecture

- $\mathcal{S}_{\text{data}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\text{data}}(\mathbf{x})$ - real samples;
- $\mathcal{S}_{\boldsymbol{\theta}} = \{\mathbf{x}_i\}_{i=1}^n \sim p_{\boldsymbol{\theta}}(\mathbf{x})$ - generated samples.

Define a binary function:

$$\mathbb{I}(\mathbf{x}, \mathcal{S}) = \begin{cases} 1, & \text{if } \exists\, \mathbf{x}' \in \mathcal{S} : \|\mathbf{x} - \mathbf{x}'\|_2 \leq \|\mathbf{x}' - \text{NN}_k(\mathbf{x}', \mathcal{S})\|_2; \\ 0, & \text{otherwise.} \end{cases}$$

$$\text{Pr}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\boldsymbol{\theta}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\text{data}}); \quad \text{Rec}(\mathcal{S}_{\text{data}}, \mathcal{S}_{\boldsymbol{\theta}}) = \frac{1}{n} \sum_{\mathbf{x} \in \mathcal{S}_{\text{data}}} \mathbb{I}(\mathbf{x}, \mathcal{S}_{\boldsymbol{\theta}}).$$



(a) True manifold (b) Approx. manifold

Embed the samples using a pretrained network (as in FID).

Kynkäänniemi T. et al. Improved precision and recall metric for assessing generative models, 2019

# Recap of Previous Lecture

## CLIP: Learn from Image–Caption Pairs

$$s_{\text{CLIP}}(\mathbf{x}, t) = \cos(\mathbf{f}_{\text{img}}(\mathbf{x}), \mathbf{f}_{\text{text}}(t)) \,.$$

**PickScore** learns human preferences; **VQAScore / GenEval / DPG-Bench** assess compositional correctness beyond CLIP similarity.

### Human Evaluation

**No automated metric is perfect.** Validate model quality with human judgments across multiple criteria. For image editing, assess:

1. **Instruction relevance:** Was the requested change made?
2. **Preservation:** Did content that should stay unchanged remain intact?
3. **Visual defects:** Were artifacts or distortions introduced?

Radford A. et al. Learning Transferable Visual Models From Natural Language Supervision, 2021 · Kirstain Y. et al. Pick-a-Pic, 2023
Lin Z. et al. Evaluating Text-to-Visual Generation with Image-to-Text Generation, 2024
Ghosh D. et al. GenEval, 2023 · Hu X. et al. ELLA (DPG-Bench), 2024 · Yandex. Alice AI ART 2.0

# Outline

01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

# Outline

01 **Langevin Dynamics**

02 Score Matching

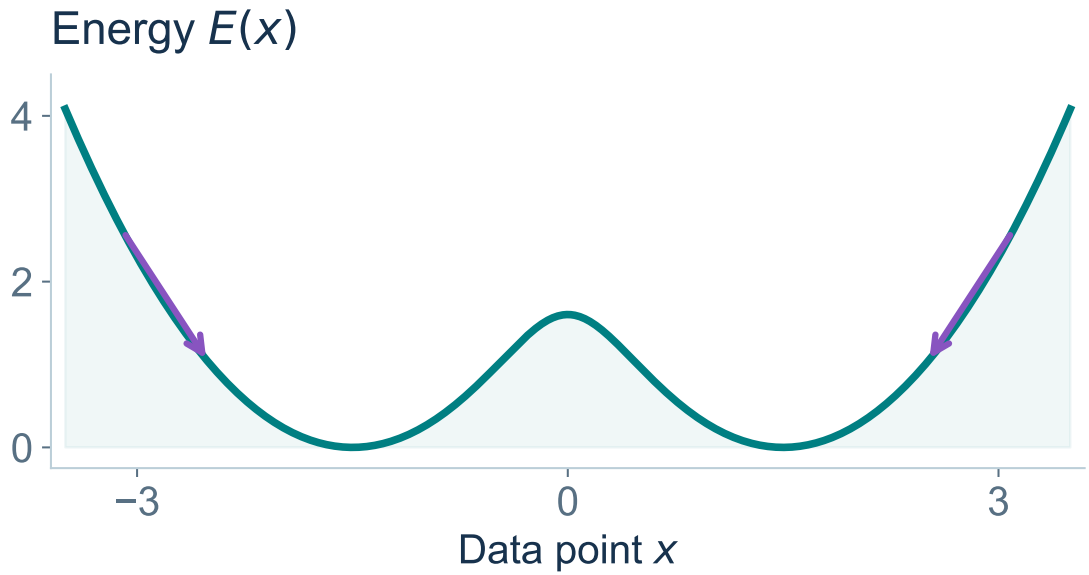
Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

# Learning an Energy Landscape

- **Autoregressive models and flows:** model a normalized density.
- **VAEs and GANs:** generate data points through latent variables.

- **An EBM assigns an energy** $E_\theta(\mathbf{x})$ to each data point $\mathbf{x}$.
- **Plausible data points have lower energy.** Several wells can represent different modes.
- Learn **relative energies**: lowering energy on training data alone is not enough.




Minimizing energy searches for a low-energy data point.
**How can we sample from the whole distribution?**

LeCun Y. et al. A Tutorial on Energy-Based Learning, 2006
Fedotov S. Energy-based models and why we might want to use them, 2026

# Energy-Based Models

## A Neural Network Defines the Energy

Use a scalar network $E_{\boldsymbol{\theta}}(\mathbf{x}) = \mathrm{NN}_{\boldsymbol{\theta}}(\mathbf{x}) \in \mathbb{R}$, with weights $\boldsymbol{\theta}$ learned from data.

$$\hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) = e^{-E_{\boldsymbol{\theta}}(\mathbf{x})} > 0, \qquad p_{\boldsymbol{\theta}}(\mathbf{x}) = \frac{\hat{p}_{\boldsymbol{\theta}}(\mathbf{x})}{Z_{\boldsymbol{\theta}}}, \qquad Z_{\boldsymbol{\theta}} = \int \hat{p}_{\boldsymbol{\theta}}(\mathbf{x}) d\mathbf{x}$$

- Lower energy means higher density; energy minima are density modes.
- We require $0 < Z_{\boldsymbol{\theta}} < \infty$, but computing it is usually intractable.

## Log-Density Gradient

The normalizing constant disappears when we differentiate with respect to $\mathbf{x}$:

$$\nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) = -\nabla_{\mathbf{x}} E_{\boldsymbol{\theta}}(\mathbf{x}) - \underbrace{\nabla_{\mathbf{x}} \log Z_{\boldsymbol{\theta}}}_{=0} = -\nabla_{\mathbf{x}} E_{\boldsymbol{\theta}}(\mathbf{x})$$

With a fixed energy model, gradient descent in $\mathbf{x}$ searches for a mode.
**Sampling must also reproduce the spread and probability mass of each mode.**

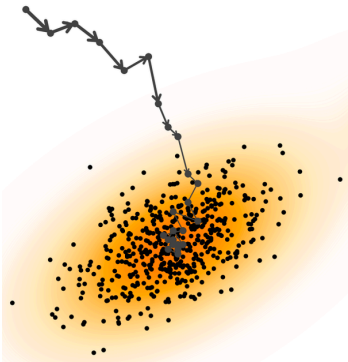
# Langevin Dynamics

## Theorem (Informal)

Starting from $\mathbf{x}_0$, the following dynamics approximately samples $p_{\boldsymbol{\theta}}$ under suitable regularity conditions, for small $\eta$ and sufficiently many steps:

$$\begin{aligned}\mathbf{x}_{l+1} &= \mathbf{x}_l + \frac{\eta}{2}\nabla_{\mathbf{x}_l}\log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta}\,\boldsymbol{\epsilon}_l \\ &= \mathbf{x}_l - \frac{\eta}{2}\nabla_{\mathbf{x}_l}E_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta}\,\boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).\end{aligned}$$

- **Without noise:** descend toward an energy minimum.
- **With noise:** explore the energy landscape.
- Update $\mathbf{x}$; keep the model parameters $\boldsymbol{\theta}$ fixed.



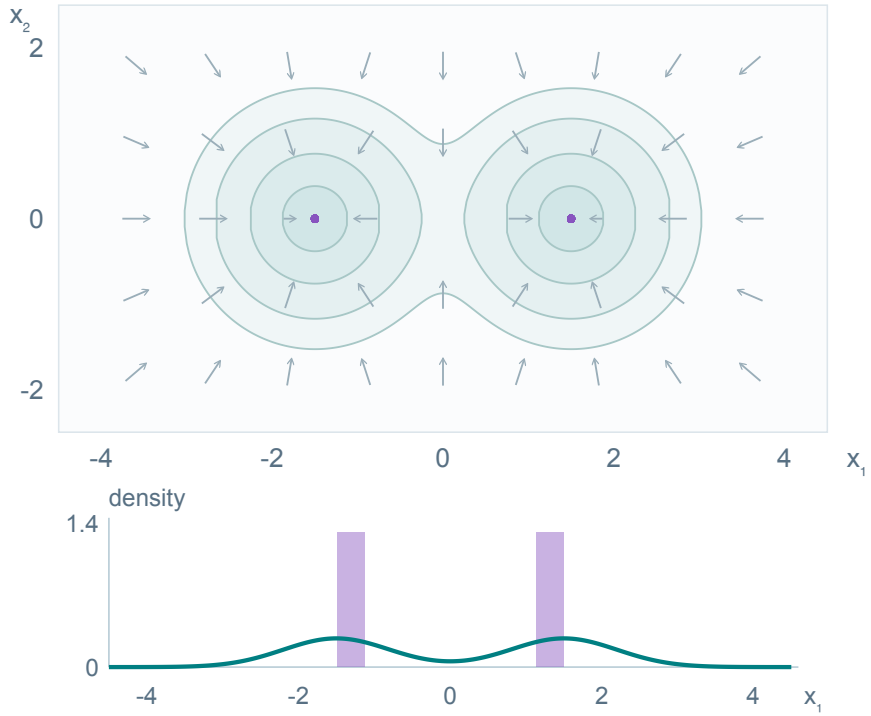
**Energy gradients suffice for approximate sampling; $Z_\theta$ is never evaluated.**

Lai C. H. et al. The principles of diffusion models, 2025.

# Langevin Dynamics: Why Add Noise?

$$p(\mathbf{x}) = \tfrac{1}{2}\mathcal{N}(\boldsymbol{\mu}_-, 0.7^2\mathbf{I}) + \tfrac{1}{2}\mathcal{N}(\boldsymbol{\mu}_+, 0.7^2\mathbf{I}), \qquad \boldsymbol{\mu}_\pm = (\pm 1.5, 0), \quad E(\mathbf{x}) = -\log p(\mathbf{x}).$$

**Noise off: descend to the minima** Step 300




Density axis adapts to the histogram.

**Noise on: sample around the wells** Step 300




Density axis adapts to the histogram.

Step size η = 0.04; finite steps and simulation time give approximate sampling.

# Outline

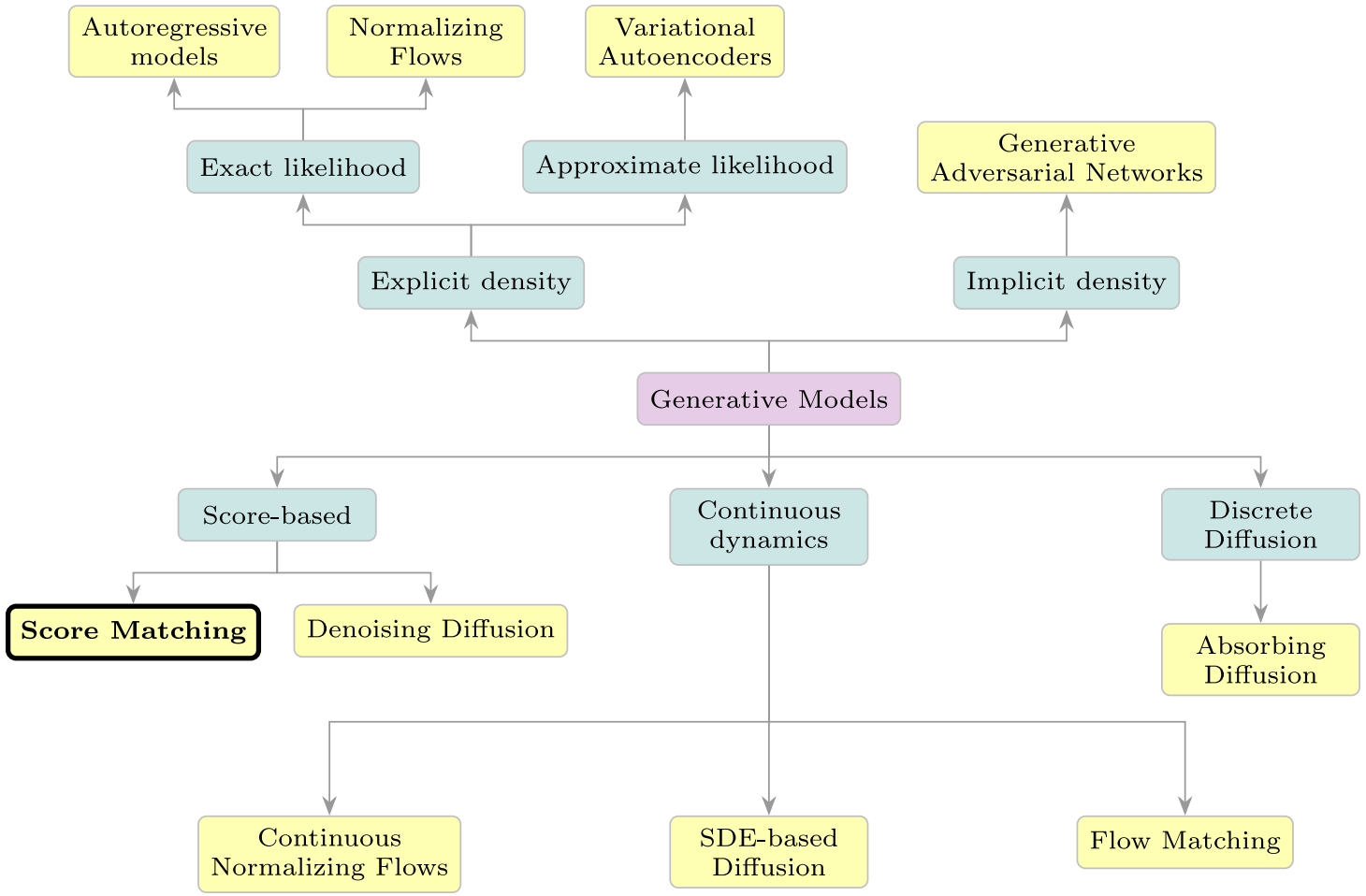
01 Langevin Dynamics

**02 Score Matching**

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

# Generative Models Taxonomy



# Score Function

## What Did We Use from the EBM?

$$\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) = -\nabla_{\mathbf{x}} E_{\boldsymbol{\theta}}(\mathbf{x})$$

Langevin dynamics uses only this vector field: the **score function**.

## Learn the Vector Field Directly

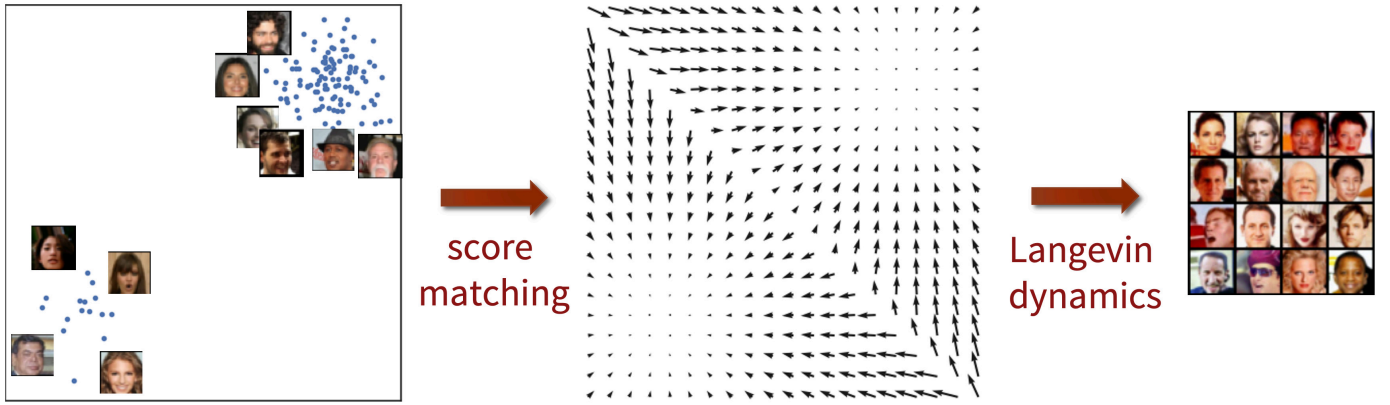
Fit a vector network $\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$, then sample:

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2}\mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta}\,\boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

A general vector network need not be the exact gradient of a scalar energy.

**How can we learn the score from data?**

# Score Matching




$$\{\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N\} \overset{\text{i.i.d.}}{\sim} p(\mathbf{x}) \qquad \mathbf{s}_\theta(\mathbf{x}) \approx \nabla_{\mathbf{x}} \log p(\mathbf{x})$$

## Fisher Divergence

$$\begin{aligned} D_F(p_{\text{data}}, p_{\boldsymbol{\theta}}) &= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \nabla_{\mathbf{x}} \log p_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \\ &= \frac{1}{2}\mathbb{E}_{p_{\text{data}}} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}} \end{aligned}$$

**Problem:** We don't know $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$.

# Why Is the Score Function Better than the Density?

## Freedom from Normalization Constants

Many distributions are defined only up to an unnormalized density $\hat{p}(\mathbf{x})$:

$$\nabla_{\mathbf{x}} \log p(\mathbf{x}) = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x}) - \underbrace{\nabla_{\mathbf{x}} \log Z}_{=0} = \nabla_{\mathbf{x}} \log \hat{p}(\mathbf{x})$$

The score function bypasses the intractable constant $Z$ entirely!

## A Complete Representation

The score $\mathbf{s}(\mathbf{x}) = \nabla_{\mathbf{x}} \log p(\mathbf{x})$ determines $\log p(\mathbf{x})$ up to the additive constant $\log p(\mathbf{x}_0)$:

$$\log p(\mathbf{x}) = \log p(\mathbf{x}_0) + \int_0^1 \mathbf{s}(\mathbf{x}_0 + t(\mathbf{x} - \mathbf{x}_0))^\top (\mathbf{x} - \mathbf{x}_0) dt$$

Modeling $\mathbf{s}(\mathbf{x})$ is as expressive as modeling $p(\mathbf{x})$ itself, while often more tractable for generative modeling.

# Outline

01 Langevin Dynamics

02 **Score Matching**

**Denoising Score Matching**
Noise-Conditioned Score Network (NCSN)

03 Forward Gaussian Diffusion Process

# Denoising Score Matching

Let us perturb the original data $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$ with Gaussian noise:

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad q(\mathbf{x}_\sigma | \mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$

$$q(\mathbf{x}_\sigma) = \int q(\mathbf{x}_\sigma | \mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x}.$$

**Assumption**

The solution to

$$\frac{1}{2} \mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

satisfies $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma) \approx \mathbf{s}_{\boldsymbol{\theta}, 0}(\mathbf{x}_0) = \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x})$ if $\sigma$ is sufficiently small.

- For small $\sigma$, the noisy-data score $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$ nearly matches the original-data score $\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})$.
- The score function $\mathbf{s}_{\boldsymbol{\theta}, \sigma}(\mathbf{x}_\sigma)$ is parameterized by $\sigma$.
- **Note:** We don't know $q(\mathbf{x}_\sigma)$, just as we don't know $p_{\text{data}}(\mathbf{x})$.

# Denoising Score Matching

We cannot evaluate the marginal score $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma)$. The theorem lets us replace it in the training objective with the conditional score $\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x})$.

## Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}$$

## Gradient of the Noise Kernel

$$\mathbf{x}_\sigma = \mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}, \qquad q(\mathbf{x}_\sigma|\mathbf{x}) = \mathcal{N}(\mathbf{x}, \sigma^2 \cdot \mathbf{I})$$
$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) = -\frac{\mathbf{x}_\sigma - \mathbf{x}}{\sigma^2} = -\frac{\boldsymbol{\epsilon}}{\sigma}$$

**Same minimizers, computable target:** train $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma)$ against $-\boldsymbol{\epsilon}/\sigma$, which we know because we added the noise ourselves.

# Denoising Score Matching: Before the Proof

## 1. Start: Learn the Data Score

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

## 2. Perturb the Data: Learn the Smoothed Score

Target: the unknown score of the noisy-data marginal $q(\mathbf{x}_\sigma)$.

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

## 3. Apply the Theorem: Same Minimizers, Known Target

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Sample $\mathbf{x}_\sigma \sim q(\mathbf{x}_\sigma|\mathbf{x})$ via $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}) - \left( -\frac{\boldsymbol{\epsilon}}{\sigma} \right) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

### After Training: Sample with the Learned Score

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

# Denoising Score Matching

## Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}.$$

## Proof

For any integrable $h$, marginalize the joint distribution:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} h(\mathbf{x}_\sigma) = \int \left( \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) h(\mathbf{x}_\sigma) d\mathbf{x}_\sigma = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} h(\mathbf{x}_\sigma).$$

Expand the squared error:

$$\begin{aligned} &\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \\ = \mathbb{E}_{q(\mathbf{x}_\sigma)} &\left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \right\|_2^2 + \underbrace{\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2}_{C_1} - 2\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[ \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right]. \end{aligned}$$

The score-norm term $C_1$ is independent of $\boldsymbol{\theta}$.

**Next:** rewrite the last cross term $-2\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[ \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right]$ using the conditional score.

# Denoising Score Matching

## Theorem

Under mild regularity conditions, the last term is independent of $\boldsymbol{\theta}$:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}.$$

## Proof (Continued)

Use marginalization and $\nabla q = q \nabla \log q$ to rewrite the cross term:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left[ \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right] = \int \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma) d\mathbf{x}_\sigma$$

$$= \int \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \left( \int q(\mathbf{x}_\sigma|\mathbf{x}) p_{\text{data}}(\mathbf{x}) d\mathbf{x} \right) d\mathbf{x}_\sigma = \iint p_{\text{data}}(\mathbf{x}) \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma|\mathbf{x}) \, d\mathbf{x} \, d\mathbf{x}_\sigma$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[ \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right].$$

Substitute and complete the square:

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 = \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left[ \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) \right\|_2^2 - 2 \mathbf{s}_{\boldsymbol{\theta},\sigma}^\top(\mathbf{x}_\sigma) \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right] + C_1$$

$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 + \text{const}.$$

# What Does Denoising Score Matching Learn?

**Why the theorem works:** MSE learns the average conditional target.

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The model sees only $\mathbf{x}_\sigma$; the target also depends on $\mathbf{x}$.
For each $\mathbf{x}_\sigma$, the MSE-optimal prediction $\mathbf{s}^*_\sigma(\mathbf{x}_\sigma)$ is the average target:

$$\mathbf{s}^*_\sigma(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x})\right]$$

## From the Optimal Prediction to the Marginal Score

$$\begin{aligned}
\mathbf{s}^*_\sigma(\mathbf{x}_\sigma) &= \int \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma|\mathbf{x}) \cdot \frac{q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})}{q(\mathbf{x}_\sigma)}d\mathbf{x} \\
&= \frac{1}{q(\mathbf{x}_\sigma)}\int \nabla_{\mathbf{x}_\sigma} q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x} \\
&= \frac{\nabla_{\mathbf{x}_\sigma}\int q(\mathbf{x}_\sigma|\mathbf{x})p_{\text{data}}(\mathbf{x})d\mathbf{x}}{q(\mathbf{x}_\sigma)} = \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma).
\end{aligned}$$

**Different conditional targets, one marginal score:** $\mathbf{s}^*_\sigma(\mathbf{x}_\sigma) = \nabla_{\mathbf{x}_\sigma}\log q(\mathbf{x}_\sigma)$.
Variation around this average adds only a model-independent constant to the loss.

# Tweedie's Formula: From Score to Predictions

For $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, with independent $\mathbf{x} \sim p_{\text{data}}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ and $\sigma > 0$, the posterior-average identity gives:

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}\left[\frac{\mathbf{x} - \mathbf{x}_\sigma}{\sigma^2}\right] = \frac{\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] - \mathbf{x}_\sigma}{\sigma^2}.$$

## Tweedie's Formula: Predict Clean Data

$$\mathbb{E}_{q(\mathbf{x}|\mathbf{x}_\sigma)}[\mathbf{x}] = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

## Predict the Added Noise

$$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{x}_\sigma] = -\sigma \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma).$$

These are MSE-optimal conditional means, not necessarily the original $\mathbf{x}$ and $\boldsymbol{\epsilon}$.

## Use the Trained Score Network

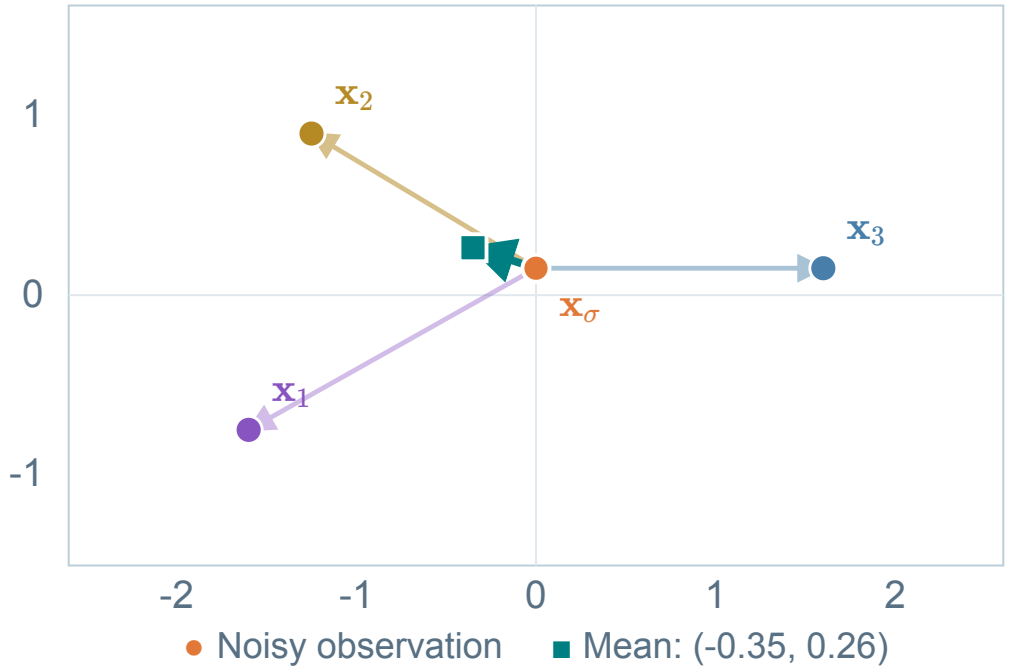
$$\hat{\mathbf{x}} = \mathbf{x}_\sigma + \sigma^2 \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma), \qquad \hat{\boldsymbol{\epsilon}} = -\sigma \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma).$$

**One score model predicts both clean data and noise**, approximating their conditional means.

# From Conditional Scores to Denoising

Toy data: three equally likely clean points $\mathbf{x}_i$, with $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ and $\sigma = 1$.

Fixed noisy observation between the clean-data modes.




## Conditional scores

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}_i) = (\mathbf{x}_i - \mathbf{x}_\sigma)/\sigma^2$$

## Bayes weights (priors 1/3 cancel)

$$w_i = q(\mathbf{x}_i|\mathbf{x}_\sigma) = \frac{\exp(-\|\mathbf{x}_\sigma - \mathbf{x}_i\|^2/(2\sigma^2))}{\sum_j \exp(-\|\mathbf{x}_\sigma - \mathbf{x}_j\|^2/(2\sigma^2))}$$

| $q(\mathbf{x}_1\|\mathbf{x}_\sigma)$ | $q(\mathbf{x}_2\|\mathbf{x}_\sigma)$ | $q(\mathbf{x}_3\|\mathbf{x}_\sigma)$ |
|---|---|---|
| 22.9% | 42.7% | 34.4% |

## Marginal score = posterior-weighted average

$$\nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \sum_i q(\mathbf{x}_i|\mathbf{x}_\sigma) \frac{\mathbf{x}_i - \mathbf{x}_\sigma}{\sigma^2}$$

## Tweedie: from the score to the clean-data mean

$$\hat{\mathbf{x}} = \mathbf{x}_\sigma + \sigma^2 \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) = \sum_i w_i \mathbf{x}_i$$

**Denoising averages plausible clean points:** $\hat{\mathbf{x}} = \sum_i w_i \mathbf{x}_i$, possibly between modes.

Vincent P. A Connection Between Score Matching and Denoising Autoencoders, 2010
Efron B. Tweedie's Formula and Selection Bias, 2011

# Denoising Score Matching: Method Overview

## 1. Start: Learn the Data Score

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta}}(\mathbf{x}) - \nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

## 2. Perturb the Data: Learn the Smoothed Score

Target: the unknown score of the noisy-data marginal $q(\mathbf{x}_\sigma)$.

$$\mathbb{E}_{q(\mathbf{x}_\sigma)} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

## 3. Apply the Theorem: Same Minimizers, Known Target

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_\sigma|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_\sigma) - \nabla_{\mathbf{x}_\sigma} \log q(\mathbf{x}_\sigma|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

Sample $\mathbf{x}_\sigma \sim q(\mathbf{x}_\sigma|\mathbf{x})$ via $\mathbf{x}_\sigma = \mathbf{x} + \sigma\boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{\boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x} + \sigma \cdot \boldsymbol{\epsilon}) - \left( -\frac{\boldsymbol{\epsilon}}{\sigma} \right) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

### After Training: Sample with the Learned Score

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta} \cdot \boldsymbol{\epsilon}_l, \qquad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I}).$$

# Outline

01 Langevin Dynamics

**02 Score Matching**

Denoising Score Matching

**Noise-Conditioned Score Network (NCSN)**

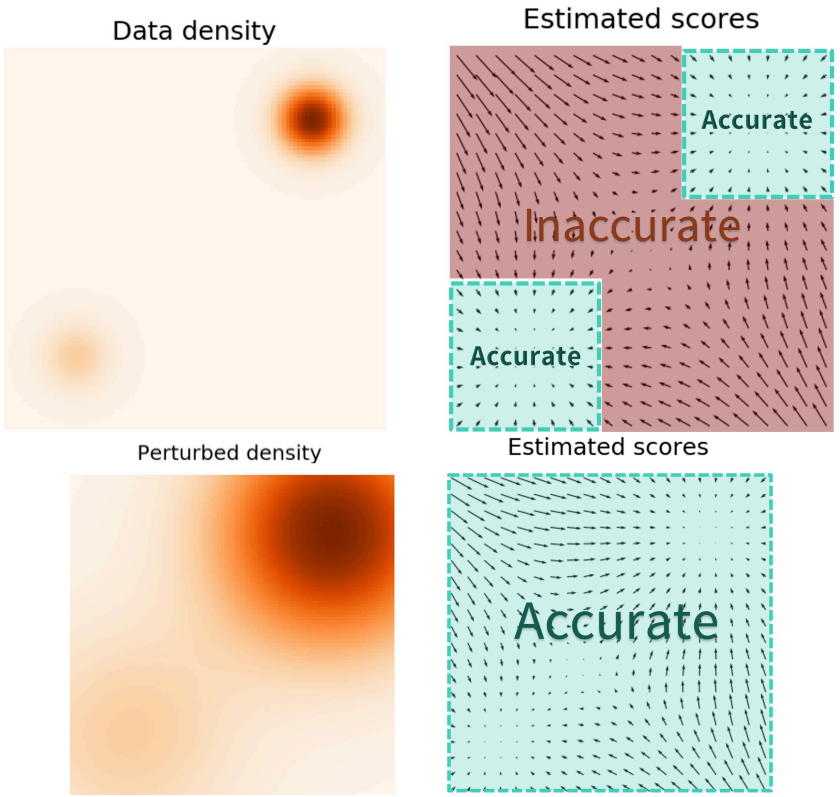
03 Forward Gaussian Diffusion Process

# Denoising Score Matching

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{\mathcal{N}(0,\mathbf{I})}\left\|\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}+\sigma\boldsymbol{\epsilon})+\frac{\boldsymbol{\epsilon}}{\sigma}\right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2}\cdot\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x}_l) + \sqrt{\eta}\cdot\boldsymbol{\epsilon}_l$$

- For **small** $\sigma$, $\mathbf{s}_{\boldsymbol{\theta},\sigma}(\mathbf{x})$ becomes inaccurate and Langevin dynamics fails to traverse modes.
- For **large** $\sigma$, robustness in low-density regions is achieved, but the model learns a distribution that is overly corrupted.

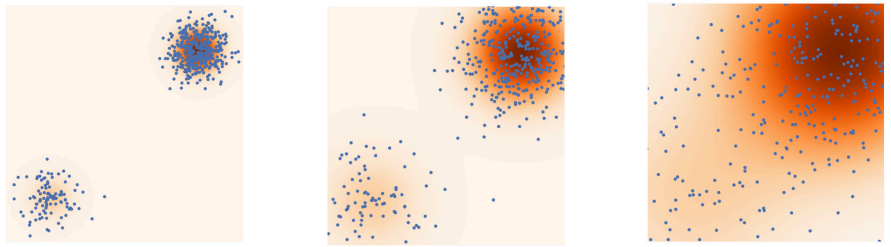


# Noise-Conditioned Score Network (NCSN)

- Specify a sequence of noise levels: $\sigma_1 < \sigma_2 < \cdots < \sigma_T$.
- Perturb each data point with different noise levels: $\mathbf{x}_t = \mathbf{x} + \sigma_t \boldsymbol{\epsilon}$, so $\mathbf{x}_t \sim q(\mathbf{x}_t)$.
- Choose $\sigma_1, \sigma_T$ such that:

$$q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \qquad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2 \mathbf{I})$$

$$\sigma_1 < \sigma_2 < \sigma_3$$





# Noise-Conditioned Score Network (NCSN)

Train the denoising score function $\mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t)$ for each noise level $\sigma_t$ using a unified weighted objective:

$$\sum_{t=1}^{T} \sigma_t^2 \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 \to \min_{\boldsymbol{\theta}}$$

The target is $-\boldsymbol{\epsilon}/\sigma_t$. The weight $\sigma_t^2$ compensates its scale:

$$\sigma_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) + \frac{\boldsymbol{\epsilon}}{\sigma_t} \right\|^2 = \left\| -\sigma_t \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) - \boldsymbol{\epsilon} \right\|^2.$$

## Training

1. Sample $\mathbf{x}_0 \sim p_{\text{data}}(\mathbf{x})$, $t \sim U\{1, T\}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$.
2. Compute noisy image $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t \boldsymbol{\epsilon}$.
3. Compute loss $\mathcal{L} = \sigma_t^2 \left\| \mathbf{s}_{\boldsymbol{\theta},\sigma_t}(\mathbf{x}_t) + \frac{\boldsymbol{\epsilon}}{\sigma_t} \right\|^2$.
4. Update $\boldsymbol{\theta}$ via stochastic gradient descent.

How do we sample from such a model?

# Noise-Conditioned Score Network (NCSN)

**Inner loop ($l$):** run Langevin dynamics at fixed $\sigma_t$, targeting $q(\mathbf{x}_t)$.
**Outer loop ($t = T, \dots, 1$):** lower the noise and reuse samples at the next level.

## Sampling (Annealed Langevin Dynamics)

1. Sample $\mathbf{x}_T^0 \sim \mathcal{N}(0, \sigma_T^2 \mathbf{I}) \approx q(\mathbf{x}_T)$.
2. Update $\mathbf{x}_t^l$ via $L$ steps of Langevin dynamics at each noise level $\sigma_t$:

$$\mathbf{x}_t^l = \mathbf{x}_t^{l-1} + \frac{\eta_t}{2} \mathbf{s}_{\boldsymbol{\theta}, \sigma_t}(\mathbf{x}_t^{l-1}) + \sqrt{\eta_t} \boldsymbol{\epsilon}_t^l.$$

3. Update $\mathbf{x}_{t-1}^0 := \mathbf{x}_t^L$ and proceed to $\sigma_{t-1}$ if $t > 1$.

## Why Gradually?

High noise helps explore different modes; lower noise restores detail. Nearby noise levels let us reuse samples instead of restarting.



# Outline

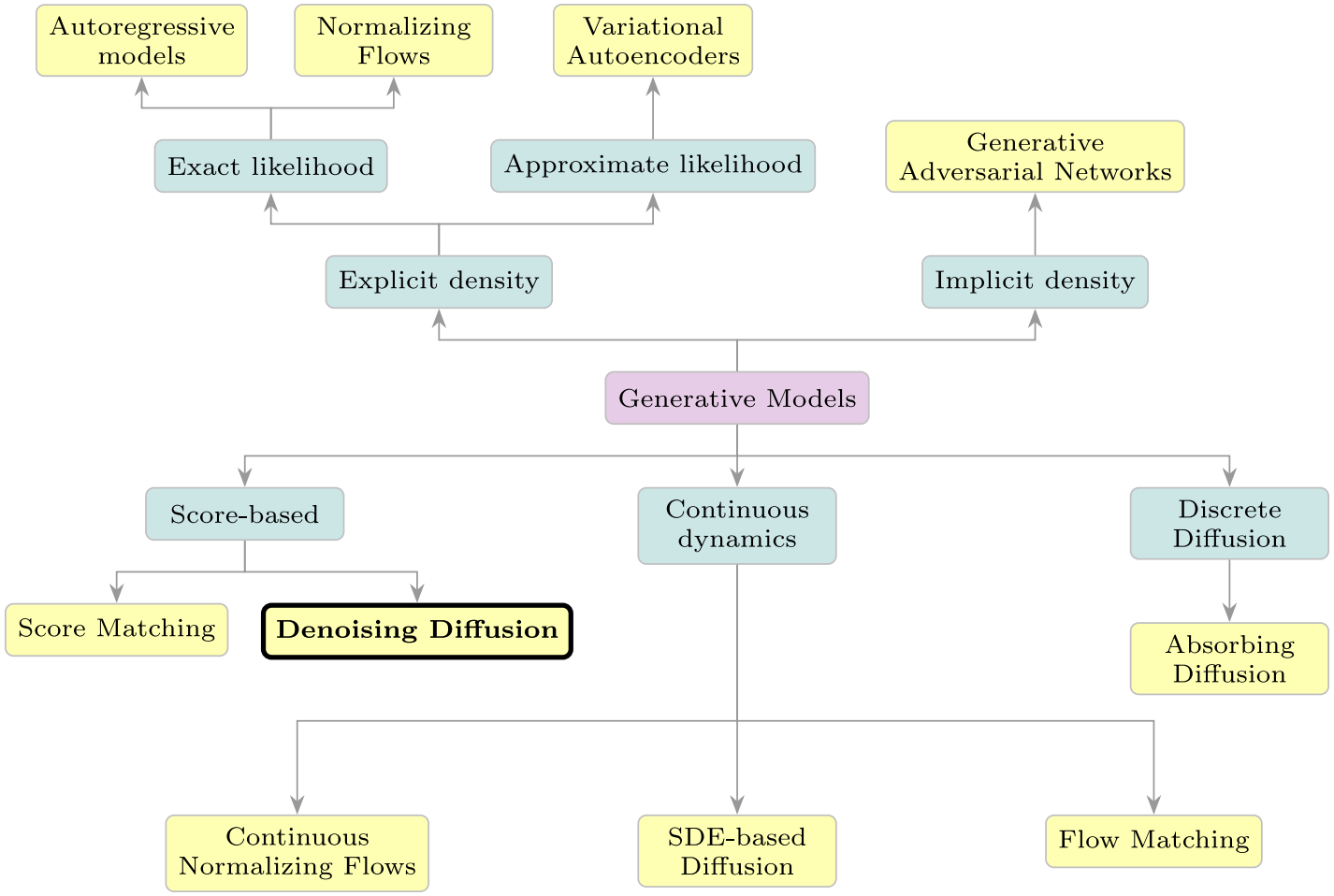
01 Langevin Dynamics

02 Score Matching

Denoising Score Matching
Noise-Conditioned Score Network (NCSN)

**03 Forward Gaussian Diffusion Process**

# Generative Models Taxonomy



# Forward Gaussian Diffusion Process

Let $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$, $\beta_t \ll 1$. Define a Markov chain:

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$$

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

## Langevin Dynamics

$$\mathbf{x}_{l+1} = \mathbf{x}_l + \frac{\eta}{2} \cdot \nabla_{\mathbf{x}_l} \log p_{\boldsymbol{\theta}}(\mathbf{x}_l) + \sqrt{\eta}\boldsymbol{\epsilon}_l, \quad \boldsymbol{\epsilon}_l \sim \mathcal{N}(0, \mathbf{I})$$

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&\approx \left(1 - \frac{\beta_t}{2}\right)\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t \\
&= \mathbf{x}_{t-1} + \frac{\beta_t}{2}(-\mathbf{x}_{t-1}) + \sqrt{\beta_t}\boldsymbol{\epsilon}_t
\end{aligned}$$

**From a Markov chain to Langevin dynamics:** for small $\beta_t$, this Gaussian chain approximates LD targeting $\mathcal{N}(0, \mathbf{I})$, with step size $\eta = \beta_t$ and Gaussian score $-\mathbf{x}_{t-1}$.

Sohl-Dickstein J. Deep Unsupervised Learning using Nonequilibrium Thermodynamics, 2015

# Forward Diffusion: Sample at Any Timestep

Fix a schedule $0 < \beta_t < 1$. The noises $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I})$ are independent across $t$ and of $\mathbf{x}_0$.

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t.$$

## Exact Conditional Distribution

Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$. Then

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}).$$

Sample $\mathbf{x}_t$ directly from $\mathbf{x}_0$, without simulating the intermediate steps:

$$\begin{aligned}
\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + (\sqrt{\alpha_t(1-\alpha_{t-1})}\boldsymbol{\epsilon}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_t) \\
&= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}'_t \\
&= \ldots = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}).
\end{aligned}$$

# When Does Only Gaussian Noise Remain?

The closed-form conditional already tells us how the signal disappears:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad \boldsymbol{\epsilon} \perp \mathbf{x}_0.$$

## The Schedule Must Remove the Signal

For $0 < \beta_t < 1$, the required condition is

$$\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s) \longrightarrow 0 \quad \Longleftrightarrow \quad \sum_{t=1}^{\infty}\beta_t = \infty.$$

Many steps alone are not enough: $\beta_t = 2^{-t}$ leaves $\bar{\alpha}_\infty > 0$.

$$\sqrt{\bar{\alpha}_t} \to 0, \qquad \sqrt{1-\bar{\alpha}_t} \to 1 \quad \Longrightarrow \quad \mathbf{x}_t \xrightarrow{d} \mathcal{N}(0, \mathbf{I}).$$

The limit no longer depends on the initial data distribution $p_{\text{data}}$.

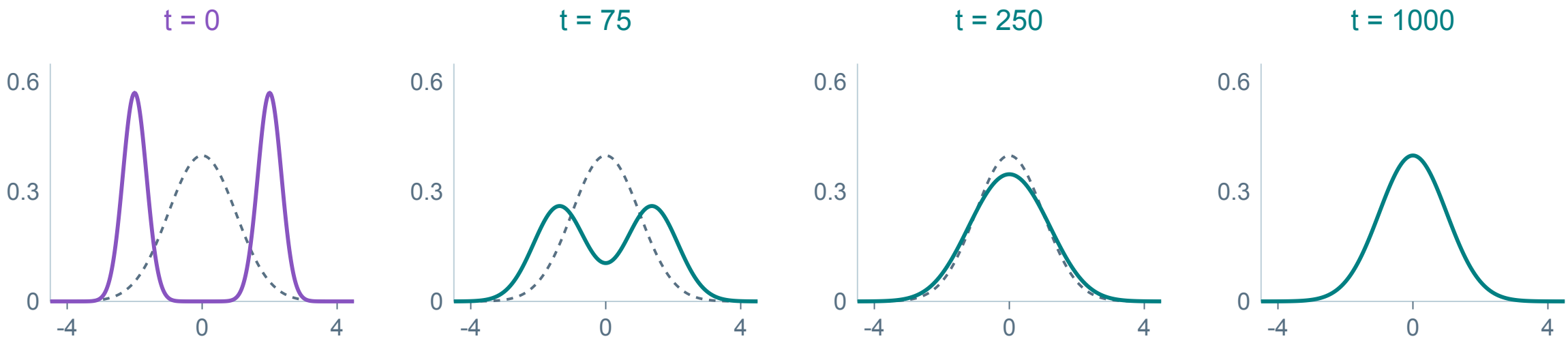
> **For finite $T$:** choose $\bar{\alpha}_T \ll 1$ with negligible residual signal $\sqrt{\bar{\alpha}_T}\mathbf{x}_0$.
> Then $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$: the terminal distribution is approximately standard Gaussian.

# Forward Diffusion: From Data to Noise

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right);$$

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right)$$

$$p_{\text{data}}(x_0) = \tfrac{1}{2}\mathcal{N}(-2, 0.35^2) + \tfrac{1}{2}\mathcal{N}(2, 0.35^2), \qquad \beta_t = 0.01$$

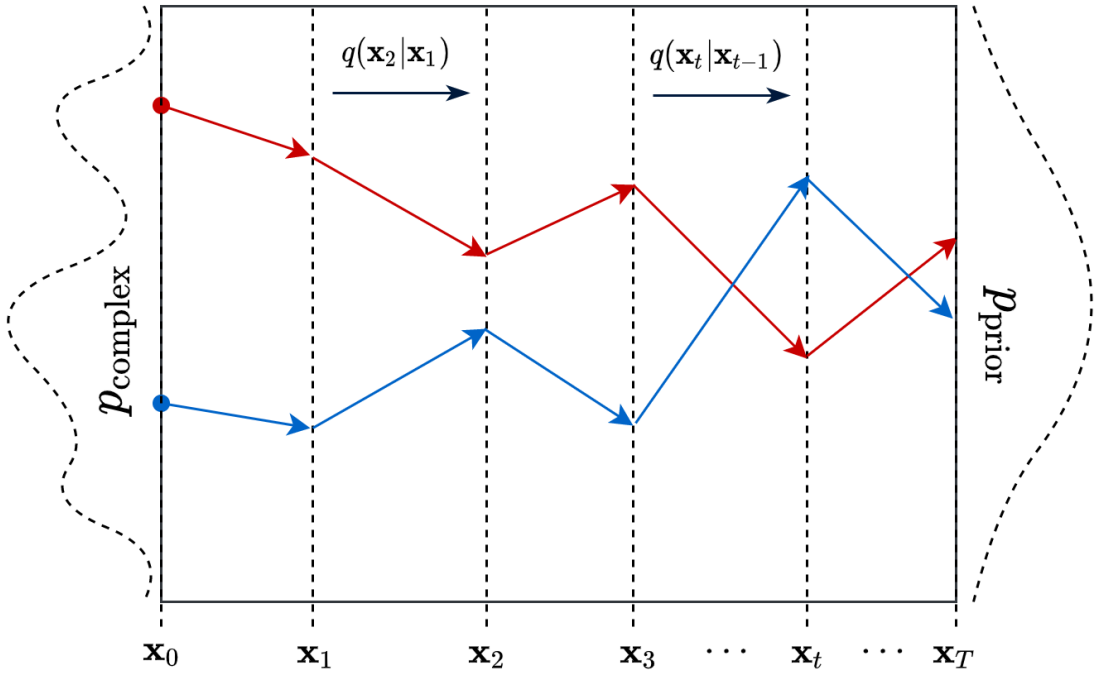



---- Standard Gaussian reference $\mathcal{N}(0,1)$ .

The plots show marginal densities; the kernels above are conditional.

# Forward Gaussian Diffusion Process

**Diffusion** describes the migration of particles from regions of high density to those of low density.




1. $\mathbf{x}_0 = \mathbf{x} \sim p_{\text{data}}(\mathbf{x})$
2. $\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\epsilon}_t,$
   $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \mathbf{I}),\ t \geq 1$
3. Choose $\bar{\alpha}_T \ll 1$ to suppress $\sqrt{\bar{\alpha}_T}\mathbf{x}_0$.
   Then $q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$.

If this process can be reversed, we can sample from $p_{\text{data}}(\mathbf{x})$ by starting from noise $\mathcal{N}(0, \mathbf{I}) \approx q(\mathbf{x}_T)$.
Our goal now becomes inverting this diffusion.

# Denoising Score Matching

## NCSN

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2\mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \sigma_T^2\mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) = -\frac{\mathbf{x}_t - \mathbf{x}}{\sigma_t^2}$$

## Gaussian Diffusion

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}), \quad q(\mathbf{x}_1) \approx p_{\text{data}}(\mathbf{x}), \quad q(\mathbf{x}_T) \approx \mathcal{N}(0, \mathbf{I})$$

$$\nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}_0) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{1-\bar{\alpha}_t}$$

## Theorem (Denoising Score Matching)

$$\mathbb{E}_{q(\mathbf{x}_t)} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t) \right\|_2^2$$
$$= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x})} \left\| \mathbf{s}_{\boldsymbol{\theta},t}(\mathbf{x}_t) - \nabla_{\mathbf{x}_t} \log q(\mathbf{x}_t|\mathbf{x}) \right\|_2^2 + \text{const}$$

**Note:** Annealed Langevin dynamics applies to diffusion, too.
NCSN with annealed Langevin dynamics is an early score-based generative framework (2019–2020).

# Summary

- EBM energy gradients and Gaussian noise give approximate Langevin samples; minimizing energy alone finds modes.
- Score matching learns the score vector field directly by minimizing Fisher divergence.
- Denoising score matching makes the score objective estimable using corrupted samples and the known corruption kernel.
- Tweedie's formula converts the score into MSE-optimal clean-data and noise predictions.
- Noise-Conditioned Score Networks balance score losses with weights $\sigma_t^2$ and sample with annealed Langevin dynamics.
- The forward Gaussian diffusion process admits a closed-form conditional distribution for any timestep.
- With a suitable noise schedule, the forward process converges to standard Gaussian noise; denoising score matching also applies to these diffusion marginals.